

CO3-AT2-CASE STUDY

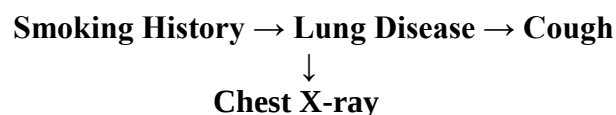
COURSE OUTCOME COVERED: CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model

Assessment Tool Weightage: 20%

Questions / Total Marks: 2 / 20

CODE OF CONDUCT

I **NAVYA SHRI A P (192312028)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



This structure allows the system to model relationships rather than treating every symptom as independent. If smoking history strongly affects the probability of lung disease, the network incorporates that dependency before interpreting the patient's cough or X-ray result.

4. Handling Uncertainty

Medical diagnosis contains uncertainty because symptoms can be shared by several diseases, measurements can be noisy, and patient histories may be incomplete. A probabilistic model does not simply return “yes” or “no”; it can provide probabilities such as:

Disease A: 0.70

Disease B: 0.20

Disease C: 0.10

This allows clinicians to understand confidence and consider additional tests when uncertainty is high. Missing observations can also be handled probabilistically instead of forcing an artificial value.

5. Role of Probability Learning

Probability learning estimates priors and conditional probabilities from historical patient data. Maximum Likelihood Estimation (MLE) can estimate parameters that make the observed training data most probable. With limited data, smoothing or Bayesian parameter estimation can reduce unreliable probability estimates.

The dataset should be divided into training, validation, and test sets. Cross-validation can also be used to estimate generalization performance.

6. Effectiveness and Real-World Constraints

The approach can be evaluated using accuracy, precision, recall, F1-score, sensitivity, specificity, ROC-AUC, and calibration of predicted probabilities. In healthcare, sensitivity is especially important for diseases where missing a positive case is dangerous.

Important real-world constraints include:

- Limited or imbalanced medical datasets.
- Missing and noisy patient records.
- Correlated symptoms and tests.
- Changes in disease prevalence across hospitals and populations.
- Privacy, security, and regulatory requirements.
- Need for interpretability and clinician oversight.
- High cost of false-negative diagnoses.

Therefore, a practical system should be used as clinical decision support rather than an unquestioned replacement for a physician.

Conclusion

A combined Bayes/Naïve Bayes/BBN approach provides a principled way to diagnose diseases under uncertainty. Naïve Bayes is computationally simple and scalable, while Bayesian Belief Networks provide a richer representation of conditional dependencies. Probability learning continuously improves parameter estimates as reliable data become available. The final system should be validated carefully on representative clinical data and should support, rather than replace, professional medical judgment.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

Proposed Spam Detection System

A spam detection system can classify each incoming email as Spam or Ham (legitimate email).

A probabilistic approach is appropriate because words, sender information, links, and other email characteristics provide uncertain evidence rather than absolute rules.

1. Naïve Bayes Classification

Let C be the class and $X = \{x_1, x_2, \dots, x_n\}$ be the extracted email features. Naïve Bayes calculates:

$$P(C | X) \propto P(C) \times \prod P(x_i | C)$$

The system first estimates the prior probabilities of Spam and Ham from labelled training emails.

It then estimates the probability of each feature occurring in each class.

Typical features include:

- Word and phrase frequencies.
- Subject-line terms.
- Presence of URLs or suspicious attachments.
- Sender/domain information.
- Number of capital letters or unusual symbols.
- TF-IDF or binary word-presence features.

For a new email, the posterior probability of each class is calculated. The class with the larger probability is selected.

Example:

$$P(\text{Spam} | \text{Email}) = 0.92$$

$$P(\text{Ham} | \text{Email}) = 0.08$$

The email is therefore classified as Spam.

Laplace/additive smoothing should be applied so that an unseen word does not make the complete probability product zero.

2. Bayes Optimal Classifier

Instead of selecting one hypothesis, the Bayes Optimal Classifier considers all hypotheses H weighted by their posterior probabilities:

$$P(c | x) = \sum P(c | x, h) P(h | D)$$

where D is the training data.

This can produce a better theoretical decision because uncertainty about the correct model is included. However, evaluating a very large hypothesis space can be computationally expensive. Naïve Bayes is therefore attractive in practical spam filtering because it is fast, simple, and easy to update.

3. Maximum Likelihood Estimation vs MDL

Maximum Likelihood Estimation (MLE) selects parameter values that maximize the likelihood of the observed training data:

$$\theta_{ML} = \arg \max_{\theta} P(D | \theta)$$

For feature selection, MLE tends to favour features that improve how well the training data are explained. With a small dataset, however, a highly flexible model may fit accidental patterns or noise.

Minimum Description Length (MDL) follows the principle of selecting a model that minimizes the total description length:

$$L(\mathbf{Model}) + L(\mathbf{Data} \mid \mathbf{Model})$$

Thus, MDL balances model complexity and the amount of information needed to encode the remaining errors. A feature is useful when its contribution to explaining the data is greater than the extra complexity it introduces.

Comparison:

MLE:

- Focuses mainly on maximizing data likelihood.
- Can fit training data very closely.
- May overfit when the hypothesis/model is too flexible.
- Does not directly penalize model complexity.

MDL:

- Balances model complexity and data fit.
- Encourages compact explanations.
- Helps control overfitting.
- Can be useful when many candidate features are available.

Therefore, MLE may obtain excellent training performance, whereas MDL can prefer a simpler feature set that generalizes better to unseen emails.

4. System Performance and Trade-offs

The system should be evaluated using a held-out test set or cross-validation. Important measures include accuracy, precision, recall, F1-score, false-positive rate, and confusion matrix.

For spam filtering, false positives are particularly important because a legitimate email incorrectly placed in spam may be missed by the user. Therefore, performance should not be judged using accuracy alone.

Model complexity creates a trade-off:

Simple model → fast, easy to train, lower computational cost, but may miss complex patterns.

Complex model → potentially captures more patterns, but requires more data, more computation, and may overfit.

A practical system can use feature selection, smoothing, regularization-like controls, and periodic retraining.

5. Effect of Sample Size

Sample size strongly affects classification accuracy. With very few labelled emails, probability estimates are unreliable and rare words may receive poor estimates. As the number of training examples increases, estimates become more stable and the classifier generally generalizes better.

A larger dataset is especially important when the feature space is large. Class imbalance should also be addressed because a dataset containing overwhelmingly more legitimate emails can bias the classifier.

6. Effect of Hypothesis Space

A finite and small hypothesis space is easier to learn from limited data. As the hypothesis space becomes larger or more complex, more training examples are required to distinguish the correct decision rule from competing hypotheses.

A very large hypothesis space can lead to overfitting: the classifier may learn accidental properties of the training emails rather than genuine spam patterns. Feature selection using MDL can reduce the effective hypothesis space and improve generalization.

Sample complexity is therefore related to both the size/complexity of the hypothesis space and the desired accuracy.

Conclusion

Naïve Bayes provides an efficient and practical foundation for spam detection, while the Bayes Optimal Classifier provides a theoretically stronger approach by averaging over hypotheses according to their posterior probabilities. MLE focuses on fitting the observed data, whereas MDL explicitly balances fit with model complexity. A sufficiently large and representative training set, careful feature selection, and evaluation using precision, recall, F1-score, and false-positive rate are essential for reliable real-world spam filtering.